

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458155.6823 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.048
Transit depth (Rp/Rs)^2	5.9e-05 +/- 0.0073 [%]
Orbital Inclination (inc)	86.03 +/- 2.9
Airmass coefficient 1 (a1)	0.83 +/- 0.5
Airmass coefficient 2 (a2)	0.14 +/- 0.12
Scatter in the residuals of the lightcurve fit is	64.19 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	6.75
Transit Duration (day)	0.076 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.048 vs 0.1368 (deviation 1.84σ)
Duration fit vs published	0.076 d vs 0.0758 d (deviation 0.01σ)
Mid-time O–C	-6.5 min
Depth SNR	0.0
Residual scatter	64.19 %

Light curve

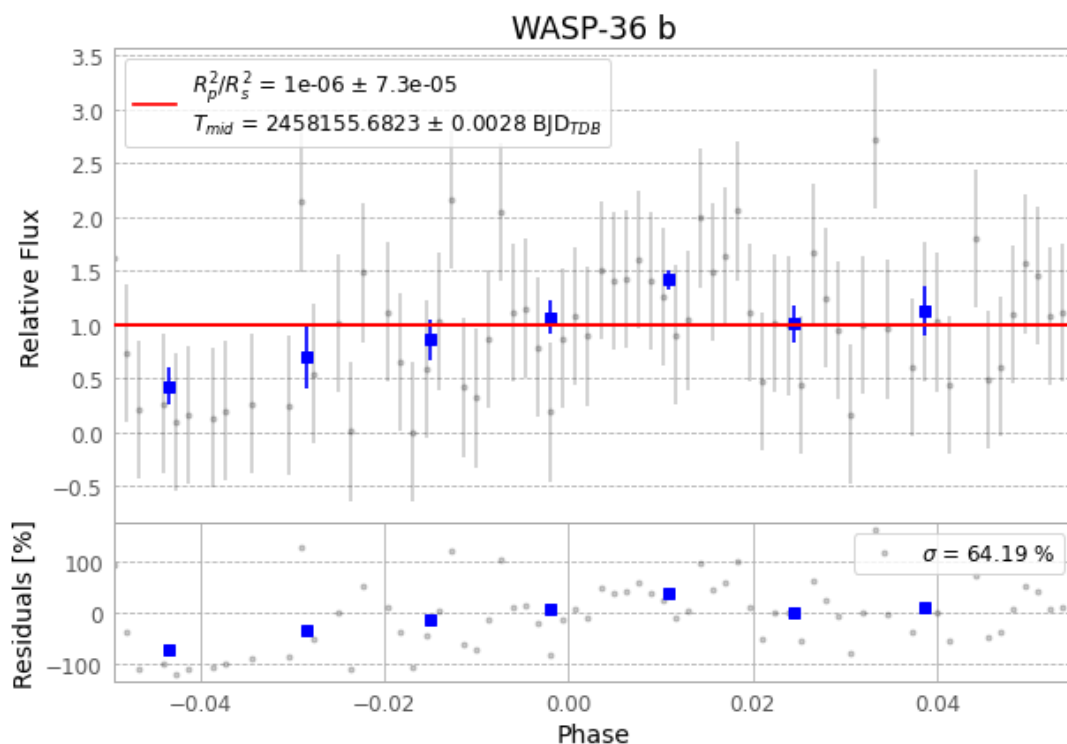


Figure 1 — FinalLightCurve_WASP-36 b_2018-02-05.png (SHA-256 66d88fc12abd4f2b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [511, 351], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 27d8292730e29840...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

78 FITS frames (51 MB), WASP-36180206022412.FITS → WASP-36180206061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-180206.log (50535c7388d5...), FinalLightCurve_WASP-36 b_2018-02-05.png (66d88fc12abd...), FinalLightCurve_WASP-36 b_2018-02-05.csv (6b47df289ea5...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 00:18Z.